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Perspective

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A call in the dark: Nocturnal flight calls and their potential to advance the study of avian migration

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ABSTRACT

Migratory birds have experienced widespread declines in abundance and face numerous threats. The conservation of migratory species relies in part on improved knowledge of active migration behavior, but this behavior is difficult to study as most birds migrate at night. Flight calls, which are species-specific calls produced by many nocturnal migrants during flight, offer an opportunity to improve our understanding of migration behavior and serve as a tool to monitor populations. Although nocturnal flight call monitoring has been historically limited to small spatial and temporal scales, recent technological advancements have allowed researchers to largely shed these constraints. Despite this expansion, there are many unanswered questions regarding the function of flight calls and the proximate drivers of calling behavior. There are also unaddressed concerns that the methods used to record nocturnal flight calls, as well as other organismal, environmental, and social factors, may bias data in ways that impede (or prohibit) comparisons across time and space. Research that addresses these limitations and potential sources of bias will advance the use of nocturnal flight call monitoring for migratory bird research.

Keywords: avian migration, bioacoustics, migration behavior, migration ecology, nocturnal flight calls

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LAY SUMMARY

- Nocturnal flight call monitoring is the primary method by which researchers can analyze active nocturnal migration at the species level.
- Researchers have been interested in nocturnal flight call monitoring for over a century; however, technological advances in recent decades have caused the field of study to expand rapidly.
- While technological advances have allowed nocturnal flight call monitoring to be used at increasingly large spatial scales and over longer time periods, many limitations and potential sources of bias need further study.
- Limitations and biases exist in our knowledge of the function of nocturnal flight calls, how calling behavior varies among species and individuals for factors such as sex, age, and weather conditions, and in the methods that we use to collect, process, and analyze acoustic recordings.
- An improved understanding of these limitations and biases will allow researchers to advance the field of avian conservation by using nocturnal flight call monitoring to its full potential.

Un llamado en la oscuridad: las llamadas en vuelo nocturnas y su potencial para avanzar en el estudio de la migración de aves

RESUMEN

Las aves migratorias han experimentado un declive generalizado en su abundancia y enfrentan numerosas amenazas. La conservación de las especies migratorias depende en parte de un mejor conocimiento del comportamiento migratorio activo, pero este comportamiento es difícil de estudiar ya que la mayoría de las aves migran durante la noche. Las llamadas en vuelo, que son llamadas específicas de cada especie producidas por muchos migrantes nocturnos durante el vuelo, ofrecen una oportunidad para mejorar nuestra comprensión del comportamiento migratorio y sirven como una herramienta para monitorear las poblaciones. Aunque el monitoreo de las llamadas en vuelo nocturnas ha estado

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históricamente limitado a escalas espaciales y temporales pequeñas, los avances tecnológicos recientes han permitido a los investigadores superar en gran medida estas limitaciones. A pesar de esta expansión, hay muchas preguntas sin respuesta sobre la función de las llamadas en vuelo y los factores que impulsan este comportamiento. También existen preocupaciones no abordadas de que los métodos utilizados para grabar las llamadas en vuelo nocturnas, así como otros aspectos de los organismos y factores ambientales y sociales, puedan sesgar los datos de formas que dificulten (o impidan) las comparaciones a lo largo del tiempo y el espacio. Los estudios que aborden estas limitaciones y posibles fuentes de sesgo permitirán avanzar en el uso del monitoreo de las llamadas en vuelo nocturnas para la investigación de las aves migratorias.

Palabras clave: bioacústica, comportamiento migratorio, ecología de la migración, llamadas en vuelo nocturnas, migración de aves

INTRODUCTION

Avian migration is a complex phenomenon involving the seasonal movements of billions of birds spanning hemispheres and connecting all regions of the world (Newton 2008, Dokter et al. 2018, Horton et al. 2019, Nussbaumer et al. 2021). Despite the number of birds in flight, the spectacle of avian migration is difficult to observe because it occurs primarily at night. While multiple methods, such as weather radar (Gauthreaux and Belser 2003, Van Doren and Horton 2018), thermal imaging (Gauthreaux and Livingston 2006, Weisshaupt et al. 2016), and moon-watching (Liechti et al. 1995, Bridge et al. 2024) have been successfully applied to study movement patterns during migration, data collected via these methods are hindered by a general inability to identify migrants to a high degree of taxonomic resolution (Horton et al. 2015a). Many species, however, broadcast species-specific calls during nocturnal movements (Farnsworth 2005), and since the late 1800s, ornithologists have listened for and counted the flight calls given by nocturnal migrants as a means to quantify migration (e.g., Libby 1899). Among the earliest work using nocturnal flight calls in migration research, Kopman (1904) cited the nocturnal flight calls of thrushes (genus *Catharus*) in mid-May in southern Louisiana as evidence of delayed spring migration, and (Tyler 1916) inferred the identity of nocturnal migrant flight calls by matching the known migration phenology of species to the temporal patterns of certain call notes. Early nocturnal flight call monitoring relied on aural counts of flight calls for data collection, while, in recent years, researchers have used autonomous recording units (ARUs; Graber and Cochran 1959). In recent decades, studies have demonstrated that counts of nocturnal flight calls correlate well with other methods to estimate the abundance of migrants at a given location and across entire regions while also allowing for the identification of migrants at species resolution (Evans and Mellinger 1999, Farnsworth et al. 2004, Horton et al. 2015a, Van Doren et al. 2024, Osterhaus et al. 2024). Today, the detection and classification of nocturnal flight calls is the primary method by which researchers can characterize migrant passage at finer taxonomic resolution during nocturnal migratory flights at both small and large scales.

Over the last 2 decades, the number of studies using nocturnal flight call monitoring has increased greatly (Figure 1). While part of this increase may be attributed to an increase in general scientific activity (Bornmann et al. 2021), it is likely that some portion of this proliferation can be attributed to the heightened interest in, and use of, nocturnal flight call monitoring as the tools and technologies needed for this method have advanced. Despite this increased research attention, many fundamental questions remain unanswered regarding the function of nocturnal flight calls, including our ability to identify all calls to the species level, and how flight calling behavior varies across time, space, species, sex, social context, and en route environmental conditions (Farnsworth 2005). If such variability exists in calling behavior, as is suggested (e.g., Tegeler et al. 2018, Gillings and Scott 2021, Ress et al. 2024), then nocturnal flight call data may contain sources of bias that are currently unaccounted for.

A review of the state of the science of nocturnal flight calls and their application in migration research has not occurred since 2005. At that time, computing capabilities as they relate to the storage and analysis of large acoustic datasets were considered the primary obstacles in nocturnal flight call research (Farnsworth 2005). With computing capabilities greatly expanded in the last 2 decades, we focus our discussion on continued limitations, the potential sources of bias noted above, and detail the role that nocturnal flight call monitoring may play in the study and conservation of avian migrants in the future. We describe current approaches to using nocturnal flight call monitoring to study bird migration, recent technological advances in these approaches, and highlight recent applications. While our discussion focuses primarily on research

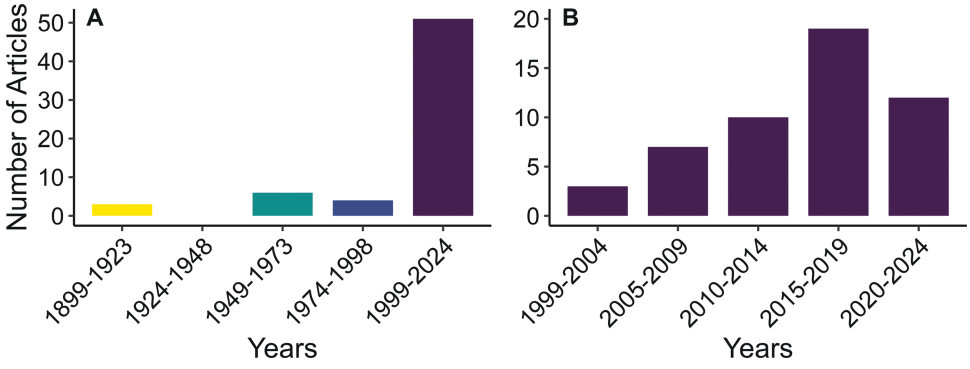


FIGURE 1. Number of peer-reviewed publications using nocturnal flight call monitoring to address ecological, behavioral, or methodological questions by year from (**A**) 1899 to 2024 and (**B**) from 1999 to 2024. Data were generated using the Web of Science online database with the search term "nocturnal flight call AND bird." The search was conducted on June 25, 2024. All titles and abstracts were screened to ensure the relevance of included articles. Articles cited within relevant articles produced by the search were screened for inclusion as well.

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conducted in the Americas and, to a lesser extent, Europe, the use of nocturnal flight calls in migration research is expanding into previously understudied regions (such as South America; Van Doren et al. 2024) and may play an important role in bird conservation at a global scale in the near future (Heiss 2018, Liu et al. 2024).

NOCTURNAL FLIGHT CALLS: WHO MAKES THEM AND WHAT DO THEY MEAN?

Nocturnal flight calls are produced during nocturnal migration and are typically single notes ranging from 50 to 300 ms in duration, occurring in the 1–9 kHz frequency range (Farnsworth 2005). In the order Passeriformes, many species are known to produce flight calls during nocturnal migration, with the notable exceptions of vireos (Vireonidae), mimids (Mimidae), Old World warblers (Sylviidae), and New World flycatchers (Tyrannidae) (Farnsworth 2005). While nocturnal flight calling behavior of species in Passeriformes has received most of the focus, flight calling has also been documented in Charadriiformes, Pelecaniformes, Cuculiformes, and other orders (La 2012).

While nocturnal flight calling is widespread in species around the globe, it seems to be most prevalent in New World species, with fewer Old World species known to produce nocturnal flight calls (Farnsworth 2005). To date, nocturnal flight calls have received the most research attention in the American and Afro-European flyways. However, even in these regions where nocturnal flight calls have been well researched, there are still species with calls that are poorly known or yet to be identified (Evans et al. 2017, Evans 2022). In other regions, such as the East Asian–Australasian Flyway, nocturnal flight call monitoring is in its early stages, but is already providing valuable insights into nocturnal bird migration within these regions (Liu et al. 2024).

As reviewed by Farnsworth (2005), the function of flight calls during nocturnal migration is not fully understood, but it is thought to be related to the stimulation of migratory restlessness or the maintenance of groups in migration. Strong evidence for this conclusion comes from early studies of nocturnal flight call function, showing that caged birds increased activity in response to the playback of recorded flight calls during migration (Hamilton 1962). Thake (1981) concluded that nocturnal flight calls may facilitate orientation and coordinate movements, and this conclusion was recently supported by Winger et al. (2019), who found evidence for flight calls possibly acting as a "social attractant" during migration, and Van Doren et al. (2025) who found that flight calls may maintain social connections during nocturnal migration. More recently, analyses of variability in individual nocturnal flight calling behavior and alteration of calling in response to conspecifics have suggested that calling may be used to share information (such as age, sex, and individual identity) between conspecifics during migratory flights (Morris et al. 2016, Griffiths et al. 2016, Tegeler et al. 2018, Ress et al. 2024).

Variability in the acoustic features of nocturnal flight calls would allow for the sharing of information among individuals and for many calls to be identified to species. However, while species-specific identification is possible for most calls, certain species groups (such as the "zeep" complex of warblers [Parulidae], which includes 9 species [Sanders and Mennill 2014]), produce calls that exhibit little interspecific variability. The lack of variation in calls from this group of warblers has historically hindered our ability to assign species-level identifications, although recent advances in bioacoustic approaches are improving our ability to differentiate among members of this group (Keen et al. 2014, Landsborough et al. 2019).

RECENT APPLICATIONS

In recent years, nocturnal flight calls have been used to answer an array of ecological and methodological questions, ranging from small-scale analyses of flock dynamics to large-scale monitoring of migration throughout a migratory season (Figure 2). For example, research examining call rates in relation to light pollution has found higher rates in areas with increased light at night (Watson et al. 2016, Gillings and Scott 2021), which may in turn cause increased collisions with lighted structures by calling species (Winger et al. 2019). Other behavioral studies have used nocturnal flight calls to address questions related to flock composition and dynamics. Gayk and Mennill (2020, 2023) used a 3-dimensional array of ARUs on the shore of Lake Michigan and found that flocks are typically composed of species that make similar-sounding calls. In terms of migration ecology, flight calls have been used to assess drivers of migration behavior and how birds move across the landscape. Changes in call rates of nocturnal migrants have been associated with weather variables including increasing tailwinds, crosswinds, cloud cover, and humidity (Hüppop and Hilgerloh 2012, Smith et al. 2014). In the northeastern United States, an array of 9 ARUs was used in combination with machine-listening software to monitor nocturnal flight calls across an entire fall season and track migration passage at the species level, finding a close correspondence to passage estimates from weather radar and citizen science (eBird) data (Van Doren et al. 2023). Lastly, nocturnal flight calls from single locations, or from many locations, have been used as points of comparison to other methods, such as radar and mist-netting, for studying migration. These studies have shown positive correlations between estimates of migration traffic from nocturnal flight call monitoring and other methods and have demonstrated how the combination of methods allows for inferences that are not possible if a method is used in isolation (Gagnon et al. 2010, Sanders and Mennill 2014, Horton et al. 2015b, Blake et al. 2023, Van Doren et al. 2023, Weisshaupt et al. 2024, Osterhaus et al. 2024).

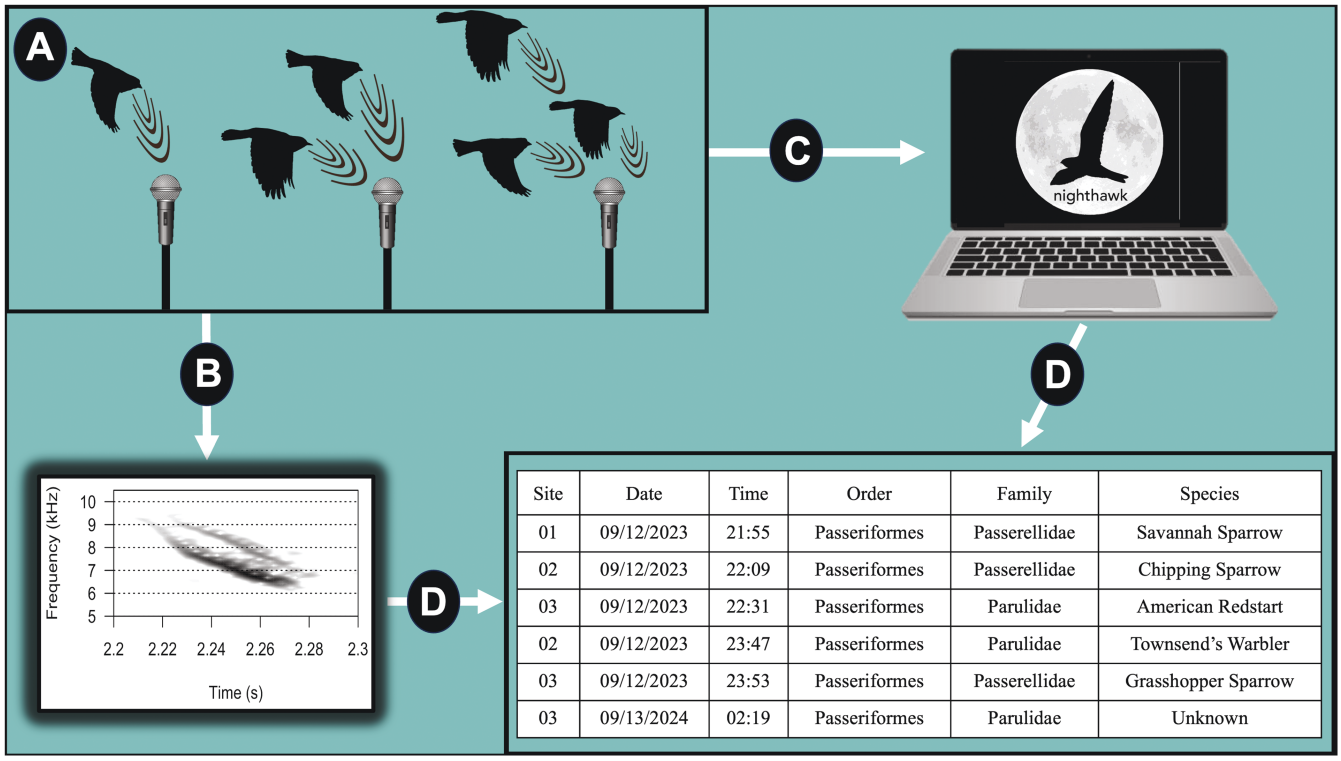
RECENT ADVANCES IN METHODOLOGY

Early researchers used nocturnal flight calls to document migration passage and examine drivers of migration, but these studies were limited in spatial and temporal scale by the available technologies (Farnsworth 2005). Recent advancements in recorder technology (Gibb et al. 2019) and computing methods and capacity (Lostanlen et al. 2019, Van Doren et al. 2024) have allowed the use of nocturnal call monitoring as a tool to study migration over large spatial and temporal applications (Van Doren et al. 2023, 2024).

Autonomous Recording Units

Advancements in ARU technology in recent years have included extended battery life, increased storage, improved

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Site	Date	Time	Order	Family	Species
01	09/12/2023	21:55	Passeriformes	Passerellidae	Savannah Sparrow
02	09/12/2023	22:09	Passeriformes	Passerellidae	Chipping Sparrow
03	09/12/2023	22:31	Passeriformes	Parulidae	American Redstart
02	09/12/2023	23:47	Passeriformes	Parulidae	Townsend's Warbler
03	09/12/2023	23:53	Passeriformes	Passerellidae	Grasshopper Sparrow
03	09/13/2024	02:19	Passeriformes	Parulidae	Unknown

FIGURE 2. A schematic representing typical data collection and processing methods for studies using nocturnal flight call data where (**A**) nocturnal flight calls are recorded at multiple locations using autonomous recording units; (**B**) collected recordings can be manually screened using visual and aural methods; however, this can be very time intensive; or (**C**) recordings can be processed in an automated fashion using detector/classifier programs such as Nighthawk; and (D) data can be tabulated by temporal or taxonomic area of interest as pursuant to the questions under study.

durability, and enhanced recorder performance (Gibb et al. 2019). In parallel, the costs of ARUs and data storage devices have declined (Hill et al. 2018, Sethi et al. 2018). These improvements have allowed researchers to employ a greater number of ARUs and collect high-quality data across larger spatial scales. With increased interest in the acoustic monitoring of environments (Sutherland et al. 2016, Sugai et al. 2019), it is likely that recording technologies will continue to improve and costs decrease.

Automated Detection and Classification of Flight Calls

Another major factor limiting early nocturnal flight call studies was the lack of automated detection and classification systems, which required researchers to manually screen recordings to gather flight call data. However, over the past 2 decades, automated systems have been developed that allow for tens of thousands of hours of acoustic recordings to be processed in a few hours (Lostanlen et al. 2019, Van Doren et al. 2024). Important precursors that paved the way for the development of these automated systems were programs that enhanced the ability of bioacousticians to view, process, and analyze sound data (such as *Raven* [K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology 2024] and *Audacity* [Audacity Team 2024]), and breakthroughs in machine learning processes used in nocturnal flight call detection/classification (Koops et al. 2014, Lostanlen et al. 2019).

At present, *Nighthawk* is the state-of-the-art automated detector/classifier for nocturnal flight calls in recordings. It performs well in detecting and classifying nocturnal flight calls from recordings across the Americas and can be trained to specific regions using relatively small training datasets (~50–100 hr of annotated recordings) (Van Doren et al. 2024). Other programs, such as *BirdVoxDetect*, are still in use and have provided important steps in the advancement of automated flight call identification (Lostanlen et al. 2022). Recent work by Lostanlen et al. (2024) highlights the novel end-to-end pipeline and workflow implemented by *BirdVoxDetect*, which may serve as a template for future iterations of automated nocturnal flight call detectors and classifiers.

Three-Dimensional Arrays

One of the most exciting innovations in the study of nocturnal flight calls has been the use of 3-dimensional recording arrays to sample individual calls from multiple microphones simultaneously. Arrays allow for the triangulation of the position of migrants and 3-dimensional reconstructions of flock dynamics during migration. While the techniques critical to the application of 3-dimensional arrays were developed in other animal systems (Spiesberger and Fristrup 1990, Clark and Ellison 2000, Blumstein et al. 2011), the first study to use a 3-dimensional array to monitor birds during migration was conducted by Stepanian et al. (2016) and served as a proof of concept and guide for further studies. Gayk and Mennill (2020) used a recording array to assess the ability and accuracy of the array to triangulate warbler flight calls, finding variability in triangulation accuracy among species, likely due to differences in call frequency and amplitude. With the same array, Gayk and Mennill (2023) analyzed calling behavior and flock composition in migrating birds, finding

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that species with similar-sounding calls often flocked together during the early diurnal hours. While arrays hold significant promise, their usage is computationally intensive, requires accurately synchronized hardware, and the detection range is limited to a small sampling range (as is true for nocturnal flight call monitoring in general, see below). As technologies for recording and triangulating calls improve, 3-dimensional arrays will likely have a larger role in answering questions related to the function of nocturnal flight calls, as well as illuminating previously unknown flock dynamics and behaviors of migrating birds.

CURRENT LIMITATIONS AND POTENTIAL BIASES

Despite the increasingly widespread use of nocturnal flight call monitoring, persistent limitations remain, and potential sources of bias stemming from methodological, organismal/individual (e.g., individual identity, sex, and species), environmental (e.g., weather conditions), and social factors are largely unaccounted for and poorly understood. Two decades ago, Farnsworth (2005) identified the following as key areas needing research: (1) understanding the function and proximate causes of flight calls and (2) factors that influence variability in calling. While subsequent studies have addressed these questions (Adams et al. 2012, Morris et al. 2016, Griffiths et al. 2016, Ress et al. 2024), the extent of recent research has been surprisingly limited given the impact these findings could have on inferences drawn from flight call monitoring. Here, we discuss each of these factors as potential limitations and biasing factors in the use of nocturnal flight call monitoring to study avian migration, particularly in quantifying migrant passage. It is important to note that these same factors may offer powerful tools for studying the behavioral ecology of birds in active nocturnal migration.

Organismal, Environmental, and Social Effects on Flight Call Behavior

Flight calls represent not only the presence of an individual bird but also the behavioral state of that individual. Variation in individuals' calling rates may be substantial and can be influenced by a variety of factors such as age, sex, body condition, or other aspects of life history or ecology (Morris et al. 2016). Call rates in response to playback of a call have been shown to vary among individuals within species (Morris et al. 2016), between age classes (Tegeler et al. 2018), and between sexes (Tegeler et al. 2018). For example, if juveniles of some species call more frequently than adults (Tegeler et al. 2018), studies analyzing call data in fall migration may overestimate passage rates later in the migration season if juvenile birds migrate later than adults in those species.

Environmental factors may contribute to variation in calling behavior as well. Multiple studies have found that call rates vary with weather (Hüppop and Hilgerloh 2012, Smith et al. 2014, Van Doren et al. 2023) and anthropogenic lighting (Watson et al. 2016, Gillings and Scott 2021). Other anthropogenic alterations to the environment, such as noise pollution, could potentially impact flight calling, leading to changes in acoustic characteristics and calling rates (Slabbekoorn and Peet 2003, Dowling et al. 2012, Derryberry et al. 2020). Furthermore, for studies examining nightly or seasonal passage, it is important to know how calling behavior varies within the timespan of a night or across a migratory season for individuals or species and what drives this variability. Multiple studies have found that call rates increase throughout the night, perhaps because of migrant flight height or habitat searching behavior (Farnsworth et al. 2004, Horton et al. 2015a, Osterhaus et al. 2024). Different groups of species also show different within-night calling trends (Van Doren et al. 2014), but whether this variation directly reflects species-specific differences in vocalization patterns or indirectly reflects differences in migration behavior (e.g., differential timing of migratory flights or differential selection of flight altitudes) is unclear.

In addition to environmental factors, social context may impact calling behavior. In a captive experiment (Lanzone et al. 2009), warblers were more likely to call when they heard the calls of other migrants and rarely called spontaneously (Morris et al. 2016). Additionally, individuals are more likely to call in response to conspecific calls than to heterospecifics (Morris et al. 2016), lending support to the hypothesis that flight calls play a role in communication with related migrants. This work highlights the fact that call detections and the estimates of migration traffic that researchers derive from these counts of detections are driven by more than just the absolute number of migrants flying over a sampling location. An improved understanding of the factors that influence calling behavior would allow researchers to control for the biases in flight call data stemming from these factors (as Van Doren et al. [2023] did for weather), further improving the ability of researchers to make robust inferences from call data.

Methodological Limitations and Biases

In addition to understanding and accounting for the potential organismal, environmental, and social factors that may alter calling behavior, it is important to understand the biases and limitations of the sampling approaches used to collect flight call data. For instance, Horton et al. (2015b) analyzed the attenuation of calls under varying levels of temperature and humidity, finding that detectability varied substantially across these conditions, with calls at 0°C and 5% relative humidity having ~80% more energy at 150 m than the same call at 20°C and 50% relative humidity. Attenuation may also occur at variable rates across recording sites based on surrounding vegetative or anthropogenic structure (Royle 2018), and background noises from wind, rainfall, insects, or human-generated sources also impact sound detection. Any form of attenuation or masking, whether it be by natural or anthropogenic means, will bias call counts unless the attenuation is at consistent levels throughout the duration of the recording period and stable among recording locations, which is unlikely for most sources.

Similar to attenuation, the height at which migrants fly in relation to the height of where recordings are being collected may vary within and among nights, as well as within and among species and across studies (Dokter et al. 2010, Horton et al. 2015a). The maximum height of detection of flight calls by ground-based recorders has not been well defined but is generally assumed to be within a few hundred meters of ground level (Evans and Mellinger 1999, Evans and Rosenberg 2000, Horton et al. 2015b), which is lower than the average flight height of many migrants (350–650 m above ground level) (La Sorte et al. 2015, Horton et al. 2016). Therefore, call counts are biased toward species that typically

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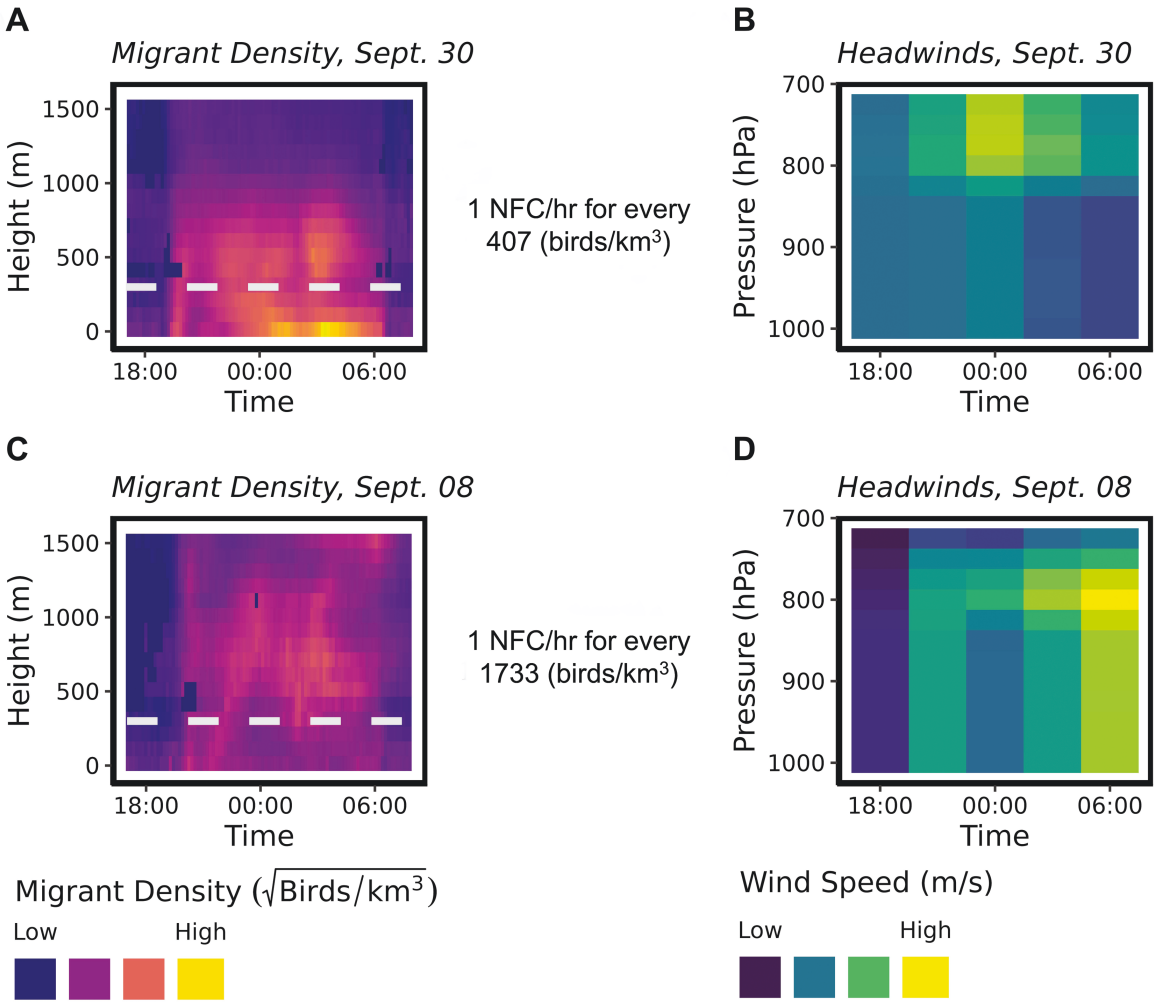


FIGURE 3. An example of how weather conditions (in this case, strong headwinds aloft) may lower the flight height of migrants, causing an increase in call rates (nocturnal flight call per hour, NFC/hr) detected at ground-based autonomous recording units that is not attributable to changes in the density of birds aloft (as measured by radar). Panel (**A**) depicts avian migrant density at various heights above ground level (agl) as measured by the KHDX Next Generation Weather Radar in southern New Mexico on the night of September 30, 2022, showing migrants clustering at low flight heights, most of which are below the horizontal dotted line representing the maximum sampling height of ground-based autonomous recording units (although this height has not been well defined). Panel (**B**) depicts headwinds aloft (assuming southbound migration) within the area sampled by the KHDX radar on the same night, indicating strong headwinds aloft and weaker headwinds at ground level. Panels (**C**) and (**D**) depict the same data as panels (**A**) and (**B**) but for a night (September 8, 2022) in which headwinds were more evenly distributed, resulting in more migrants flying above recorder sampling range. Across both nights, avian migrant density ranged (per pixel) from 0 to 12.8 $\left(\sqrt{\text{birds/km}^3}\right)$ and headwind speeds ranged from 3.06 to 7.03 m s^{-1}. At this location, the 1,000 hPa pressure level corresponds with ~50–100 m agl and the 700 hPa pressure level corresponds to ~3,000 m agl. Radar data were generated using the *bioRad* package in R (Dokter et al. 2019), wind data were accessed from the North American Regional Reanalysis dataset (Mesinger et al. 2006), and nocturnal flight call recordings were collected from an array of autonomous recording units near the radar site.

migrate at a lower height above ground level. Also, on nights in which atmospheric conditions (such as winds aloft) lower the average flight height of migrants, flight call detections may increase simply because more birds will be flying within range of the recorder, not because of an increase in total migrant passage (Figure 3). Further complicating the issue, each different species may adjust migration height differently in response to atmospheric conditions, which would necessitate species-specific information to correct for biases in flight call data attributed to variation in flight height, a computational task of daunting complexity.

At a more fundamental level, biases may be present in recordings and resulting call counts from variability in the microphones or recording settings used (see Adams et al. 2012, Darras et al. 2020), or stemming from how data are handled from study to study (MacPhail et al. 2024). The sampling range of different microphone and recorder models has not been well quantified (and would vary considerably under various environmental conditions), which creates difficulty when authors attempt to define the scale at which they are collecting recordings. Further compounding this issue, calling amplitude likely varies between species, leading to species-to-species differences in sampling range. Despite calls for studies into the sampling range and performance of microphones currently on the market, little is known about how data collected by different recording setups may compare (Gibb et al. 2019). Additionally, given the large amounts of data (many terabytes for some studies) that can be collected by nocturnal flight call monitoring, compression of audio files to reduce data storage requirements is sometimes used. Whether or not the data are compressed for storage, and what type of compression is used, can lead to variation in performance of human

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listeners and automated detectors/classifiers when analyzing the acoustic data, potentially limiting our ability to compare output between datasets with different data handling protocols (Araya-Salas et al. 2017, MacPhail et al. 2024).

Finally, our inability to identify calls to the species level for certain groups of migrants using current methods limits our ability to make species-level inferences for these groups from flight call data. As noted by Landsborough et al. (2019), our ability to identify flight calls to the species level may be of particular importance for species of greatest conservation concern, such as *Parkesia motacilla* (Louisiana Waterthrush) and *Setophaga cerulea* (Cerulean Warbler), both of which are currently lumped within the "zeep" complex with 7 other warbler species (Landsborough et al. 2019). Additionally, even for species with distinct calls, we are not always able to accurately identify all calls to species. While recent automated detection and classification algorithms (such as *Nighthawk*; Van Doren et al. 2024) have improved in the classification of flight calls at the species level, none of these algorithms are perfect, and there is still room for further improvement. Furthermore, while automated detectors and classifiers have increased the ability of researchers to process nocturnal flight call recordings and extract species identifications, manual identification of calls by a human observer still plays an important role in creating and fine-tuning these algorithms and nocturnal flight call research (including citizen science) in general. Even though multiple resources are available that can aid individuals in learning the identification of flight calls for some species (such as Flight Calls of Migratory Birds [Evans and O'Brien 2002], the Macaulay Library (https://www.macaulaylibrary.org/), The Sound Approach—Nocturnal Flight Calls [Constantine et al. 2019], and xeno-canto [https://xeno-canto.org/]), no single guide to flight calls has been produced to serve this function. Such a guide would be indispensable to researchers and citizen scientists, but also difficult to produce as nocturnal flight calls are still poorly characterized for some species.

FLIGHT CALL MONITORING TO ADVANCE AVIAN CONSERVATION

Nocturnal flight call monitoring is an important tool with the potential to transform the way we study and monitor migration. Perhaps the most beneficial aspect of nocturnal flight call monitoring over other methods used to document migration is the ability to identify species. If we can better understand and account for the various limitations and biases related to nocturnal flight call monitoring, there is much that stands to be gained for the advancement of avian conservation—especially as we work towards comprehensive continental-scale monitoring.

Open-source automated nocturnal flight call detector/classifiers would provide a cost-effective method to collect species-specific data at large scales. These tools, in combination with low-cost ARUs, would allow researchers to obtain high-quality data at relatively low costs. Already, the sampling of migration at continental scales using weather radar has produced important findings and insights into the status of migratory bird populations and their responses to global climate change. If we can apply nocturnal flight call monitoring at similar scales, the species specificity of this technique would allow researchers to address the status and trends of specific species at continental scales. The examination of how migrants respond to drought, wildfires, extreme weather, and other forms of environmental variability would also benefit from such large-scale data collection.

Nocturnal flight call monitoring can be used to track the passage of migratory species through a region with high levels of temporal resolution (Van Doren et al. 2023, 2024). Year-to-year comparisons of passage timing using this technique at large scales may be a valuable method to track plasticity, or lack thereof, of migrants in response to changes in seasonality and primary productivity driven by climate change (Robertson et al. 2024). In addition to migration timing, researchers could pursue questions pertaining to the abundance, productivity, or overwinter survival of populations at large scales, similar to Dokter et al. (2010) using radar data, but with the addition of species specificity. As research is beginning to show that nocturnal flight calls and calling behavior may encode age (Tegeler et al. 2018), sex (Griffiths et al. 2016), and potentially individual identity (Ress et al. 2024), it is possible that we may be able to glean even more information from nocturnal flight call monitoring in the near future.

Nocturnal flight call monitoring has also been a useful method in documenting migration patterns of rare or secretive species that are difficult to observe through other methods. For example, a study focused on nocturnal flight calls of migrating grassland birds documented the passage of *Laterallus jamaicensis* (Black Rail) through a region where the species was not previously known to migrate or overwinter (Evans and Mellinger 1999). Nocturnal flight call monitoring has also been demonstrated to perform better than more typical survey methodologies (i.e., citizen science data collected through eBird; Sullivan et al. 2009) in documenting the passage of secretive migrants such as *Hylocichla mustelina* (Wood Thrush) and *Catharus fuscescens* (Veery) (Van Doren et al. 2024). The expansion of nocturnal flight call monitoring to landscape-level coverage and into previously understudied regions may reveal previously unknown migration routes and aid in the study of rare and secretive species, providing invaluable information for conservation.

The species specificity of nocturnal flight call monitoring allows for unique insights into migration ecology that cannot be gained from other monitoring methods alone. However, when used on its own, data collected by and inferences made from nocturnal flight call monitoring may be limited by the relatively small spatial and altitudinal sampling range of individual recorders (e.g., relative to an individual weather radar site). Therefore, the contemporaneous sampling of migration using multiple methodologies offers a more complete understanding of migration. The combination of monitoring methodologies has been demonstrated by many studies (Horton et al. 2015a, Strand et al. 2023, Van Doren et al. 2023, Giuntini et al. 2024, Osterhaus et al. 2024), and will allow for insights into these complex movements that extend beyond what is possible for a single method.

FUTURE DIRECTIONS

Regardless of the unresolved limitations and biases discussed herein, nocturnal flight call monitoring in its current state is a valuable method for studying migration and provides a meaningful measure of migration activity that is consistent

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with data from Doppler weather radars and citizen science observations (Van Doren et al. 2023, 2024). Recognizing its limitations and addressing its biases will further increase its usefulness, allowing researchers to advance the field of avian conservation, as detailed above, using nocturnal flight call monitoring. Priority topics for future study include further examination of organismal, environmental, and social factors that may impact calling behavior, as well as investigation of variables leading to biases in sampling of nocturnal flight calls including variability in recording setups and attenuation by a variety of sources. In these future endeavors, nocturnal flight call researchers must work collaboratively to establish standard protocols for the collection (e.g., ensuring that different microphones and recording setups are sampling nocturnal flight calls similarly) and processing (e.g., establishing best practices for data compression and storage) of acoustic recordings. Standardization of these methods would improve our ability to make broad generalizations using recordings collected by multiple studies.

Multiple methods are available to further analyze the factors impacting calling behavior. Experiments with captive birds have proven vital to the study of organismal traits and their effect on calling behavior, and pairing these studies with the experimental manipulation of the subjects' environment may shed light on the impact of environmental factors. The study of wild birds in active migration using 3-dimensional microphone arrays will be important in increasing our understanding of the function of nocturnal flight calls, the behavioral underpinnings that lead to calling variability, and how calls may be used in a social context. Furthermore, pairing acoustic monitoring with other survey methods shows great promise, not only for providing high-quality data but also for generating an enhanced understanding of the underlying limitations and biases within nocturnal flight call data. As nocturnal flight call monitoring expands to a continental scale, efforts should be made to place flight call monitoring locations in association with other migration monitoring tools, such as weather radar and MOTUS stations. Finally, as the size and cost of microphones decrease, the onboard recording of calls from individuals in migration is likely to become available for more researchers and a greater diversity of migrants (Gill et al. 2016, Stowell et al. 2017). Onboard recordings from individuals during migration, especially if paired with data from other wearable sensors, will allow for a much more thorough examination of how organismal, environmental, and social factors may impact calling behaviors and calling rates.

Finally, researchers must continue to explore new applications for nocturnal flight call monitoring to advance avian conservation. For instance, while currently unexplored, it is possible that nocturnal flight call data collected immediately after sunset or immediately prior to sunrise could provide meaningful information about stopover habitat usage with species-level resolution. Therefore, future studies should assess how nocturnal flight call data collected within these periods correspond with stopover densities as detected by weather radar.

CONCLUSION

Migratory birds are under threat from many factors including accelerating climate change and a multitude of anthropogenic stressors, which have led to widespread declines in abundance (Rosenberg et al. 2019). To protect and conserve migratory bird populations, which are critical to the healthy functioning of many of the world's ecosystems (Bauer and Hoye 2014), and to pursue solutions to the grand challenges facing migration ecology (Bauer et al. 2019), researchers must be able to use the methods at their disposal to their fullest extent. Species-specific data collected at the large scales at which migration occurs will be indispensable in pursuing these answers. Nocturnal flight call monitoring can provide this data and will contribute greatly to these pursuits as the method is further refined. Furthermore, when applied contemporaneously with other methods, nocturnal flight call monitoring can allow for deeper insights into migration ecology and behavior than can be accomplished with any one method alone.

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Ethics statement

This study did not involve the capture or handling of animals, and no behaviors or actions were altered by the methods used.

Conflict of interest statement

The authors have no conflicts of interest to declare.

Author contributions

D.M.O. and T.F.W. conceived the idea of this manuscript. M.J.D., P.L.C., D.M.O., and T.F.W. obtained funding for the project. D.M.O. wrote the first draft of the manuscript, with input from T.F.W. and M.J.D. All authors provided input on manuscript content, and reviewed, and edited the manuscript.

Data Availability

Analyses reported in this article can be reproduced using the data and code provided by Osterhaus et al. (2025).

LITERATURE CITED

Adams, A. M., M. K. Jantzen, R. M. Hamilton, and M. B. Fenton (2012). Do you hear what I hear? Implications of detector selection for acoustic monitoring of bats. *Methods in Ecology and Evolution* 3:992–998.

Araya-Salas, M., G. Smith-Vidaurre, and M. Webster (2017). Assessing the effect of sound file compression and background noise on measures of acoustic signal structure. *Bioacoustics* 28:57–73.

Downloaded from https://academic.oup.com/auk/article/142/2/ukaf002/7951360 by Purdue University user on 08 July 2026

Audacity Team (2024). Audacity: Free audio editor, recorder, music making and more! https://www.audacityteam.org/.

Bauer, S., and B. J. Hoye (2014). Migratory animals couple biodiversity and ecosystem functioning worldwide. *Science* 344:1242552.

Bauer, S., J. Shamoun-Baranes, C. Nilsson, A. Farnsworth, J. F. Kelly, D. R. Reynolds, A. M. Dokter, J. F. Krauel, L. B. Petterson, K. G. Horton, and J. W. Chapman (2019). The grand challenges of migration ecology that radar aeroecology can help answer. *Ecography* 42:861–875.

Blake, M., M. Fylling, D. Leick, T. Rodriguez, and K. Stone (2023). Evaluating two avian monitoring tools for detecting relative abundance of species during autumn migration. *Journal of Field Ornithology* 94:art4.

Blumstein, D. T., D. J. Mennill, P. Clemins, L. Girod, K. Yao, G. Patricelli, J. L. Deppe, A. H. Krakauer, C. Clark, K. A. Cortopassi, S. F. Hanser, et al. (2011). Acoustic monitoring in terrestrial environments using microphone arrays: Applications, technological considerations and prospectus. *Journal of Applied Ecology* 48:758–767.

Bornmann, L., R. Haunschild, and R. Mutz (2021). Growth rates of modern science: A latent piecewise growth curve approach to model publication numbers from established and new literature databases. *Humanities and Social Sciences Communications* 8:1–15.

Bridge, E. S., W. T. Honeycutt, A. J. Chen, R. Miller, and J. F. Kelly (2024). Social behavior among nocturnally migrating birds revealed by automated moonwatching. *Ornithology* 141:ukad055.

Clark, C. W., and W. T. Ellison (2000). Calibration and comparison of the acoustic location methods used during the spring migration of the bowhead whale, *Balaena mysticetus*, off Pt. Barrow, Alaska, 1984–1993. *The Journal of the Acoustical Society of America* 107:3509–3517.

Constantine, M., M. Robb, and A. van den Berg (2019). The Sound Approach - Nocturnal Flight Calls. *The Sound Approach*. https://soundapproach.co.uk/night-flight-calls/.

Darras, K. F. A., F. Deppe, Y. Fabian, A. P. Kartono, A. Angulo, B. Kolbrek, Y. A. Mulyani, and D. M. Prawiradilaga (2020). High microphone signal-to-noise ratio enhances acoustic sampling of wildlife. *PeerJ* 8:e9955.

Derryberry, E. P., J. N. Phillips, G. E. Derryberry, M. J. Blum, and D. Luther (2020). Singing in a silent spring: Birds respond to a half-century soundscape reversion during the COVID-19 shutdown. *Science* 370:575–579.

Dokter, A. M., P. Desmet, J. H. Spaaks, S. Van Hoey, L. Veen, L. Verlinden, C. Nilsson, G. Haase, H. Leijnse, A. Farnsworth, W. Bouten, and J. Shamoun-Baranes (2019). bioRad: Biological analysis and visualization of weather radar data. *Ecography* 42:852–860.

Dokter, A. M., A. Farnsworth, D. Fink, V. Ruiz-Gutierrez, W. M. Hochachka, F. A. La Sorte, O. J. Robinson, K. V. Rosenberg, and S. Kelling (2018). Seasonal abundance and survival of North America's migratory avifauna determined by weather radar. *Nature Ecology & Evolution* 2:1603–1609.

Dokter, A. M., F. Liechti, H. Stark, L. Delobbe, P. Tabary, and I. Holleman (2010). Bird migration flight altitudes studied by a network of operational weather radars. *Journal of The Royal Society Interface* 8:30–43.

Dowling, J. L., D. A. Luther, and P. P. Marra (2012). Comparative effects of urban development and anthropogenic noise on bird songs. *Behavioral Ecology* 23:201–209.

Evans, R., and D. K. Mellinger (1999). Monitoring grassland birds in nocturnal migration. In *Ecology and Conservation of Grassland Birds of the Western Hemisphere* (P. D. Vickery and J. R. Herkert, Editors). *Studies in Avian Biology* 19:219–229.

Evans, W. (2022). Presumed nocturnal flight call of the Green-tailed Towhee. *Western Birds* 53:143–152.

Evans, W. R., M. Grosselet, G. Ruiz Michael, W. R. Evans, M. Grosselet, and G. Ruiz Michael (2017). An unidentified nocturnal flight call from southern Mexico. *Huitzil* 18:131–140.

Evans, W. R., and M. O'Brien (2002). *Flight Calls of Migratory Birds Eastern North American Landbirds*. http://oldbird.org/pubs/fcmb/start.htm.

Evans, W. R., and K. V. Rosenberg (2000). Acoustic monitoring of night-migrating birds: A progress report. *Acoustic Monitoring of Night-Migrating Birds: A Progress Report*: 151–159.

Farnsworth, A. (2005). Flight calls and their value for future ornithological studies and conservation research. *The Auk* 122:733–746.

Farnsworth, A., S. A. Gauthreaux Jr, and D. van Blaricom (2004). A comparison of nocturnal call counts of migrating birds and reflectivity measurements on Doppler radar. *Journal of Avian Biology* 35:365–369.

Gagnon, F., M. Bélisle, J. Ibarzabal, P. Vaillancourt, and J.-P. L. Savard (2010). A comparison between nocturnal aural counts of passerines and radar reflectivity from a Canadian weather surveillance radar. *The Auk* 127:119–128.

Gauthreaux, Jr, S. A., and J. W. Livingston (2006). Monitoring bird migration with a fixed-beam radar and a thermal-imaging camera. *Journal of Field Ornithology* 77:319–328.

Gauthreaux, S. A., and C. G. Belser (2003). Radar ornithology and biological conservation. *The Auk* 120:266–277.

Gayk, Z. G., and D. J. Mennill (2020). Pinpointing the position of flying songbirds with a wireless microphone array: Three-dimensional triangulation of warblers on the wing. *Bioacoustics* 29:375–386.

Gayk, Z. G., and D. J. Mennill (2023). Acoustic similarity of flight calls corresponds with the composition and structure of mixed-species flocks of migrating birds: Evidence from a three-dimensional microphone array. *Philosophical Transactions of the Royal Society B: Biological Sciences* 378:20220114.

Gibb, R., E. Browning, P. Glover-Kapfer, and K. E. Jones (2019). Emerging opportunities and challenges for passive acoustics in ecological assessment and monitoring. *Methods in Ecology and Evolution* 10:169–185.

Gill, L. F., P. B. D'Amelio, N. M. Adreani, H. Sagunsky, M. C. Gahr, and A. ter Maat (2016). A minimum-impact, flexible tool to study vocal communication of small animals with precise individual-level resolution. *Methods in Ecology and Evolution* 7:1349–1358.

Gillings, S., and C. Scott (2021). Nocturnal flight calling behaviour of thrushes in relation to artificial light at night. *Ibis* 163:1379–1393.

Giuntini, S., J. Saari, A. Martinoli, D. G. Preatoni, B. Haest, B. Schmid, and N. Weisshaupt (2024). Quantifying nocturnal thrush migration using sensor data fusion between acoustics and vertical-looking radar. *Remote Sensing in Ecology and Conservation* 10:743–754.

Graber, R. R., and W. W. Cochran (1959). An audio technique for the study of nocturnal migration of birds. *The Wilson Bulletin* 71:220–236.

Griffiths, E. T., S. C. Keen, M. Lanzone, and A. Farnsworth (2016). Can nocturnal flight calls of the migrating songbird, American Redstart, encode sexual dimorphism and individual identity? *PLoS One* 11:e0156578.

Hamilton, W. J. (1962). Evidence concerning the function of nocturnal call notes of migratory birds. *The Condor* 64:390–401.

Heiss, M. (2018). Nocturnal bird migration at Besh Barmag bottleneck in Azerbaijan as revealed by means of acoustic monitoring. *Bird Conservation International* 28:593–605.

Hill, A. P., P. Prince, E. Piña Covarrubias, C. P. Doncaster, J. L. Snaddon, and A. Rogers (2018). AudioMoth: Evaluation of a smart open acoustic device for monitoring biodiversity and the environment. *Methods in Ecology and Evolution* 9:1199–1211.

Horton, K. G., W. G. Shriver, and J. J. Buler (2015a). A comparison of traffic estimates of nocturnal flying animals using radar, thermal imaging, and acoustic recording. *Ecological Applications* 25:390–401.

Horton, K. G., P. M. Stepanian, C. E. Wainwright, and A. K. Tegeler (2015b). Influence of atmospheric properties on detection of wood-warbler nocturnal flight calls. *International Journal of Biometeorology* 59:1385–1394.

Horton, K. G., B. M. Van Doren, F. A. La Sorte, E. B. Cohen, H. L. Clipp, J. J. Buler, D. Fink, J. F. Kelly, and A. Farnsworth (2019). Holding steady: Little change in intensity or timing of bird migration over the Gulf of Mexico. *Global Change Biology* 25:1106–1118.

Horton, K. G., B. M. Van Doren, P. M. Stepanian, A. Farnsworth, and J. F. Kelly (2016). Where in the air? Aerial habitat use of nocturnally migrating birds. *Biology Letters* 12:20160591.

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Hüppop, O., and G. Hilgerloh (2012). Flight call rates of migrating thrushes: Effects of wind conditions, humidity and time of day at an illuminated offshore platform. *Journal of Avian Biology* 43:85–90.

K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology (2024). Raven Pro: Interactive Sound Analysis Software. https://www.ravensoundsoftware.com/.

Keen, S., J. C. Ross, E. T. Griffiths, M. Lanzone, and A. Farnsworth (2014). A comparison of similarity-based approaches in the classification of flight calls of four species of North American wood-warblers (Parulidae). *Ecological Informatics* 21:25–33.

Koops, H., J. Van Balen, F. Wiering, L. Cappellato, N. Ferro, M. Halvey, and W. Kraaij (2014). A deep neural network approach to the lifeclef 2014 bird task. *CLEF Working Notes* 1180:634–642.

Kopman, H. (1904). Bird migration phenomena in the extreme Lower Mississippi Valley. *The Auk* 21:45–50.

La Sorte, F. A., W. M. Hochachka, A. Farnsworth, D. Sheldon, B. M. Van Doren, D. Fink, and S. Kelling (2015). Seasonal changes in the altitudinal distribution of nocturnally migrating birds during autumn migration. *Royal Society Open Science* 2:150347.

La, V. (2012). Diurnal and nocturnal birds vocalize at night: A review. *The Condor* 114:245–257.

Landsborough, B. J., J. R. Foote, and D. J. Mennill (2019). Decoding the 'zeep' complex: Quantitative analysis of interspecific variation in the nocturnal flight calls of nine wood warbler species (*Parulidae* spp.). *Bioacoustics* 28:555–574.

Lanzone, M., E. DeLeon, L. Grove, and A. Farnsworth (2009). Revealing undocumented or poorly known flight calls of warblers (Parulidae) using a novel method of recording birds in captivity. *The Auk* 126:511–519.

Libby, O. G. (1899). The nocturnal flight of migrating birds. *The Auk* 16:140–146.

Liechti, F., B. Bruderer, and H. Paproth (1995). Quantification of nocturnal bird migration by moonwatching: Comparison with radar and infrared observations. *Journal of Field Ornithology* 66:457–468.

Liu, S., T. Townshend, A. Farnsworth, D. Alexander, H. Tian, and F. Hua (2024). Monitoring bird migration using nocturnal flight calls on the East Asian–Australasian Flyway: A case from Beijing. *Bird Conservation International* 34:e19.

Lostanlen, V., A. Cramer, J. Salamon, A. Farnsworth, B. M. V. Doren, S. Kelling, and J. P. Bello (2022). BirdVox: Machine listening for bird migration monitoring. https://www.biorxiv.org/content/10.1101/2022.05.31.494155v1.

Lostanlen, V., A. Cramer, J. Salamon, A. Farnsworth, B. M. Van Doren, S. Kelling, and J. P. Bello (2024). BirdVoxDetect: Large-scale detection and classification of flight calls for bird migration monitoring. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 32:4134–4145.

Lostanlen, V., J. Salamon, A. Farnsworth, S. Kelling, and J. P. Bello (2019). Robust sound event detection in bioacoustic sensor networks. *PLoS One* 14:e0214168.

MacPhail, A. G., D. A. Yip, E. C. Knight, R. Hedley, M. Knaggs, J. Shonfield, E. Upham-Mills, and E. M. Bayne (2024). Audio data compression affects acoustic indices and reduces detections of birds by human listening and automated recognisers. *Bioacoustics* 33:74–90.

Mesinger, F., G. DiMego, E. Kalnay, K. Mitchell, P. C. Shafran, W. Ebisuzaki, D. Jović, J. Woollen, E. Rogers, E. H. Berbery, M. B. Ek, et al. (2006). North American regional reanalysis. *Bulletin of the American Meteorological Society* 87:343–360.

Morris, S. R., K. G. Horton, A. K. Tegeler, and M. Lanzone (2016). Individual flight-calling behaviour in wood warblers. *Animal Behaviour* 114:241–247.

Newton, I. (2008). *The Ecology of Bird Migration*. Academic Press, London, UK.

Nussbaumer, R., S. Bauer, L. Benoit, G. Mariethoz, F. Liechti, and B. Schmid (2021). Quantifying year-round nocturnal bird migration with a fluid dynamics model. *Journal of The Royal Society Interface* 18:20210194.

Osterhaus, D. M., K. C. Boland, K. G. Horton, B. M. Van Doren, P. L. Cutler, M. J. Desmond, and T. F. Wright (2025). Data from: A call in the dark: Nocturnal flight calls and their potential to advance the study of avian migration. *Ornithology* 142:ukaf002. https://doi.org/10.5061/dryad.g79cnp617 [Dataset].

Osterhaus, D. M., B. M. Van Doren, K. G. Horton, F. Abadi, T. F. Wright, and M. J. Desmond (2024). Evaluation of methods to estimate nocturnal bird migration activity: Comparison of radar and nocturnal flight call monitoring in the American West. *Ornithological Applications* 126:duae062.

Ress, E. M., A. Farnsworth, S. R. Morris, M. Lanzone, and B. M. Van Doren (2024). Magnolia Warbler flight calls demonstrate individuality and variation by season and recording location. *Ornithology* 141:ukad056.

Robertson, E. P., F. A. La Sorte, J. D. Mays, P. J. Taillie, O. J. Robinson, R. J. Ansley, T. J. O'Connell, C. A. Davis, and S. R. Loss (2024). Decoupling of bird migration from the changing phenology of spring green-up. *Proceedings of the National Academy of Sciences USA* 121:e2308433121.

Rosenberg, K. V., A. M. Dokter, P. J. Blancher, J. R. Sauer, A. C. Smith, P. A. Smith, J. C. Stanton, A. Panjabi, L. Helft, M. Parr, and P. P. Marra (2019). Decline of the North American avifauna. *Science* 366:120–124.

Royle, J. A. (2018). Modelling sound attenuation in heterogeneous environments for improved bioacoustic sampling of wildlife populations. *Methods in Ecology and Evolution* 9:1939–1947.

Sanders, C. E., and D. J. Mennill (2014). Acoustic monitoring of nocturnally migrating birds accurately assesses the timing and magnitude of migration through the Great Lakes. *The Condor: Ornithological Applications* 116:371–383.

Sethi, S. S., R. M. Ewers, N. S. Jones, C. D. L. Orme, and L. Picinali (2018). Robust, real-time and autonomous monitoring of ecosystems with an open, low-cost, networked device. *Methods in Ecology and Evolution* 9:2383–2387.

Slabbekoorn, H., and M. Peet (2003). Birds sing at a higher pitch in urban noise. *Nature* 424:267–267.

Smith, A. D., P. W. C. Paton, and S. R. McWilliams (2014). Using nocturnal flight calls to assess the fall migration of warblers and sparrows along a coastal ecological barrier. *PLoS One* 9:e92218.

Spiesberger, J. L., and K. M. Fristrup (1990). Passive localization of calling animals and sensing of their acoustic environment using acoustic tomography. *The American Naturalist* 135:107–153.

Stepanian, P. M., K. G. Horton, D. C. Hille, C. E. Wainwright, P. B. Chilson, and J. F. Kelly (2016). Extending bioacoustic monitoring of birds aloft through flight call localization with a three-dimensional microphone array. *Ecology and Evolution* 6:7039–7046.

Stowell, D., E. Benetos, and L. F. Gill (2017). On-bird sound recordings: Automatic acoustic recognition of activities and contexts. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 25:1193–1206.

Strand, A. I., E. S. Bridge, J. F. Kelly, P. M. Stepanian, D. J. Bodine, and J. R. Soto (2023). A multi-sensor array for detecting and analyzing nocturnal avian migration. *PeerJ* 11:e15622.

Sugai, L. S. M., T. S. F. Silva, J. W. Ribeiro, and D. Llusia (2019). Terrestrial passive acoustic monitoring: Review and perspectives. *BioScience* 69:15–25.

Sullivan, B. L., C. L. Wood, M. J. Iliff, R. E. Bonney, D. Fink, and S. Kelling (2009). eBird: A citizen-based bird observation network in the biological sciences. *Biological Conservation* 142:2282–2292.

Sutherland, W. J., S. Broad, J. Caine, M. Clout, L. V. Dicks, H. Doran, A. C. Entwistle, E. Fleishman, D. W. Gibbons, B. Keim, B. LeAnstey, et al. (2016). A horizon scan of global conservation issues for 2016. *Trends in Ecology & Evolution* 31:44–53.

Tegeler, A. K., K. G. Horton, and S. R. Morris (2018). Is flight-calling behaviour influenced by age, sex and/or body condition? *Animal Behaviour* 138:123–129.

Thake, M. (1981). Calling by nocturnal migrants: A device for improving orientation? *Vogelwarte* 31:111.

Downloaded from https://academic.oup.com/auk/article/142/2/ukaf002/7951360 by Purdue University user on 08 July 2026

Tyler, W. M. (1916). The call-notes of some nocturnal migrating birds. *The Auk* 33:132–141.

Van Doren, B. M., A. Farnsworth, K. Stone, D. M. Osterhaus, J. Drucker, and G. Van Horn (2024). Nighthawk: Acoustic monitoring of nocturnal bird migration in the Americas. *Methods in Ecology and Evolution* 15:329–344.

Van Doren, B. M., and K. G. Horton (2018). A continental system for forecasting bird migration. *Science* 361:1115–1118.

Van Doren, B. M., V. Lostanlen, A. Cramer, J. Salamon, A. Dokter, S. Kelling, J. P. Bello, and A. Farnsworth (2023). Automated acoustic monitoring captures timing and intensity of bird migration. *Journal of Applied Ecology* 60:433–444.

Van Doren, B. M., D. Sheldon, J. Geevarghese, W. M. Hochachka, and A. Farnsworth (2014). Autumn morning flights of migrant songbirds in the northeastern United States are linked to nocturnal migration and winds aloft. *The Auk: Ornithological Advances* 132:105–118.

Van Doren, B. M., J. G. DeSimone, J. A. Firth, F. Hillemann, Z. Gayk, E. Cohen, and A. Farnsworth (2025). Social associations across species during nocturnal bird migration. *Current Biology* 35:1–7.

Watson, M. J., D. R. Wilson, and D. J. Mennill (2016). Anthropogenic light is associated with increased vocal activity by nocturnally migrating birds. *The Condor: Ornithological Applications* 118:338–344.

Weisshaupt, N., M. Maruri, and J. Arizaga (2016). Nocturnal bird migration in the Bay of Biscay as observed by a thermal-imaging camera. *Bird Study* 63:533–542.

Weisshaupt, N., J. Saari, and J. Koistinen (2024). Evaluating the potential of bioacoustics in avian migration research by citizen science and weather radar observations. *PLoS One* 19:e0299463.

Winger, B. M., B. C. Weeks, A. Farnsworth, A. W. Jones, M. Hennen, and D. E. Willard (2019). Nocturnal flight-calling behaviour predicts vulnerability to artificial light in migratory birds. *Proceedings of the Royal Society B: Biological Sciences* 286:20190364.

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